

Hints & Solutions

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Q1 (4) - Relation between roots and coefficients

Given $x^2 + 5|x| + 4 = 0$

$$\Rightarrow (|x| + 1)(|x| + 4) = 0$$

$$|x| + 1 = 0$$

$$\Rightarrow |x| = -1$$

$$\text{or } |x| + 4 = 0$$

$$\text{or } |x| = -4$$

which is not possible.

Hence, no value of x exist.

Q2 (4) - Relation between roots and coefficients

Given, $x^2 - (a - 2)x - a - 1 = 0$

Let α, β be roots of the given equation, then

$$\alpha + \beta = a - 2 \text{ and } \alpha\beta = -a - 1$$

$$\text{Now, } \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (a - 2)^2 + 2(a + 1) = a^2 - 2a + 6 = (a - 1)^2 + 5$$

Hence, $\alpha^2 + \beta^2$ is minimum at $a = 1$

Q3 (3) - Relation between roots and coefficients

Given, α, β are roots of $x^2 - 6x - 2 = 0$

$$\therefore \alpha^2 - 6\alpha - 2 = 0$$

$$\Rightarrow \alpha^{10} - 6\alpha^9 - 2\alpha^8 = 0 \quad \dots (i)$$

$$\text{and } \beta^2 - 6\beta - 2 = 0$$

$$\Rightarrow \beta^{10} - 6\beta^9 - 2\beta^8 = 0 \quad \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(\alpha^{10} - \beta^{10}) - 6(\alpha^9 - \beta^9) - 2(\alpha^8 - \beta^8) = 0$$

$$\Rightarrow a_{10} - 2a_8 = 6a_9$$

$$\therefore \frac{a_{10} - 2a_8}{2a_9} = 3$$

Q4 (2) - Relation between roots and coefficients

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Given, $\alpha + \beta = -2k$

and $\alpha^2 + \beta^2 = 4k^2 - 2k$

Now, $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$

$\therefore 4k^2 = 4k^2 - 2k + 2\alpha\beta$

$\Rightarrow \alpha\beta = k$

Now, $S = \frac{\alpha + \beta}{\alpha} + \frac{\alpha + \beta}{\beta}$

$= \frac{(\alpha + \beta)^2}{\alpha\beta} = \frac{(-2k)^2}{k} = 4k$

and $P = \left(\frac{\alpha + \beta}{\alpha}\right) \left(\frac{\alpha + \beta}{\beta}\right) = 4k$

Hence, the equation having roots $\frac{\alpha + \beta}{\alpha}$ and $\frac{\alpha + \beta}{\beta}$ is $x^2 - 4kx + 4k = 0$

Q5 (4) - common root

Given, $(x + k)$ is a common factor of $x^2 + px + q$ and $x^2 + lx + m$

then $(-k)^2 + p(-k) + q = 0 \dots(i)$

$\Rightarrow k^2 - pk + q = 0$

and $(-k)^2 - lk + m = 0$

$\Rightarrow k^2 - lk + m = 0 \dots(ii)$

From Eqs. (i) and (ii), we get

$(l - p)k = m - q$

$k = \frac{m - q}{l - p} = \frac{q - m}{p - l}$

Q6 (2) - common root

Given, $3x^2 + ax + 1 = 0 \dots(i)$

and $2x^2 + bx + 1 = 0 \dots(ii)$

Let α be a common root of Eqs. (i) and (ii), then

$3\alpha^2 + a\alpha + 1 = 0$

and $2\alpha^2 + b\alpha + 1 = 0$

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Using cross-multiplication,

$$\frac{\alpha^2}{a-b} = \frac{\alpha}{2-3} = \frac{1}{3b-2a}$$

$$\Rightarrow -(a-b) = -\frac{1}{3b-2a}$$

$$\Rightarrow 3ab - 3b^2 - 2a^2 + 2ab = 1$$

$$\Rightarrow 5ab - 2a^2 - 3b^2 = 1$$

Q7 (2) - common root

Given, $ax^2 + bx + c = 0 \dots(i)$

and $x^3 + 3x^2 + 3x + 2 = 0$

$$\Rightarrow (x+2)(x^2+x+1) = 0$$

Eqs. (i) and (ii) have two common roots

i.e. $ax^2 + bx + c = 0$ is identical to

$$x^2 + x + 1 = 0$$

i.e. $a = b = c$

Q8 (3) - common root

Given, $x^2 + ax + b = 0 \dots(i)$

and $x^2 + bx + a = 0 \dots(ii)$

Eqs. (i) and (ii) have a common root, say α , then

$$\alpha^2 + a\alpha + b = 0$$

and $\alpha^2 + b\alpha + a = 0$

$$\therefore \frac{\alpha^2}{a^2-b^2} = \frac{\alpha}{b-a} = \frac{1}{b-a}$$

$$\Rightarrow -(a+b) = 1 \Rightarrow a+b = -1$$

Q9 (1) - graphs for quadratic

If $ax^2 + bx + c = 0$ does not have real and distinct roots, then

$$af(1) > 0$$

$$\Rightarrow a(a+b+c) > 0$$

Q10 (2) - graphs for quadratic

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If $ax^2 + bx + 10 = 0$ does not have real and distinct roots, then

$$f(0)f(-5) = 10(25a - 5b + 10) \geq 0$$

$$\Rightarrow 50(5a - b + 2) \geq 0$$

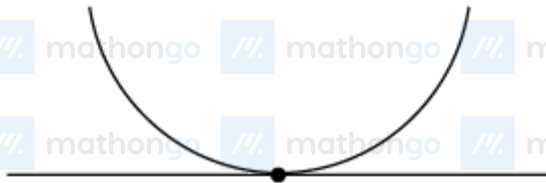
$$5a - b \geq -2$$

i.e. minimum value of $5a - b = -2$

Q11 (3) - graphs for quadratic

$f(x) = ax^2 + bx + c = 0$ such that

$$c = \frac{b^2}{4a} \Rightarrow b^2 - 4ac = 0$$



$$\left(\frac{-b}{2a}, 0 \right)$$

Equal roots at $x = \frac{-b}{2a}$

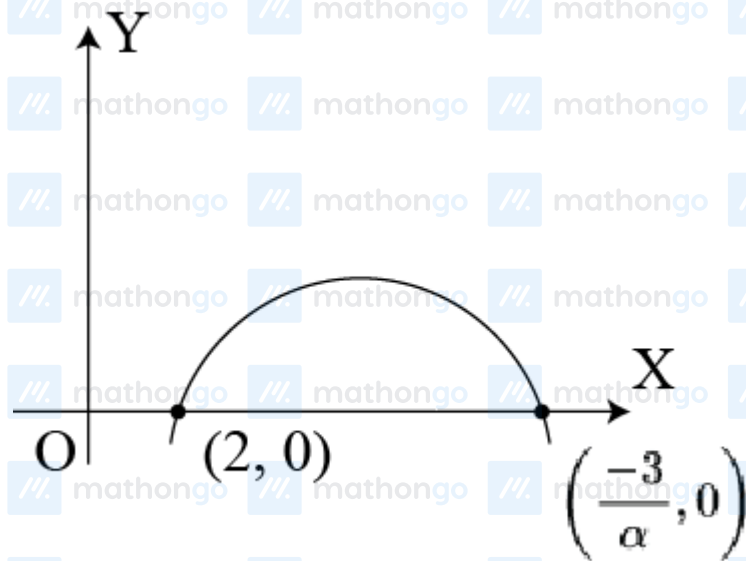
i.e. X-axis will be tangent to $f(x)$.

Q12 (4) - graphs for quadratic

Since, expression is positive for two integral values of x so the parabola $y = \alpha x^2 + |2\alpha - 3|x - 6$ will open downward

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$$\Rightarrow \alpha < 0$$

$$\text{and } y = \alpha x^2 + (3 - 2\alpha)x - 6$$

$$\Rightarrow \alpha < 0$$

$$\text{and } y = (\alpha x + 3)(x - 2)$$

$$\Rightarrow 4 < \frac{-3}{\alpha} \leq 5$$

$$\Rightarrow \frac{-3}{4} < \alpha \leq \frac{-3}{5}$$

Hence, no integral value of α exists.

Q13 (4) - graphs for quadratic

If $\min(2x^2 - ax + 2) > \max(b - 1 + 2x - x^2)$, then

$$-\frac{(a^2-16)}{8} > -\frac{[4+4(b-1)]}{-4}$$

$$\Rightarrow -a^2 + 16 > 8b$$

$$\Rightarrow a^2 + 8b < 16$$

Now, for equation $2x^2 + ax + 12 - b = 0$

$$D = a^2 - 8(12 - b) = a^2 - 96 + 8b < 16 - 96 = -80$$

i.e. $D < 0$. Hence, roots of Eq. (i) are imaginary.

Q14 (1) - Range of quadratic expressions

$$\text{Let } y = \frac{3x^2+9x+17}{3x^2+9x+7}$$

$$\Rightarrow y(3x^2 + 9x + 7) = 3x^2 + 9x + 17$$

$$\Rightarrow 3x^2y + 9xy + 7y - 3x^2 - 9x - 17 = 0$$

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$$\Rightarrow 3x^2(y-1) + 9x(y-1) + 7y - 17 = 0$$

$$\Rightarrow 3(y-1)x^2 + 9(y-1)x + (7y-17) = 0$$

which is a quadratic equation in x .

For x to be real, $D \geq 0$

$$\Rightarrow \{9(y-1)\}^2 - 4 \times \{3(y-1)\}(7y-17) \geq 0 \quad [\because D = b^2 - 4ac]$$

$$\Rightarrow 81(y-1)^2 - 12(y-1)(7y-17) \geq 0$$

$$\Rightarrow 81(y^2 + 1 - 2y) - 12\{7y^2 - 17y - 7y + 17\} \geq 0$$

$$\Rightarrow 81y^2 + 81 - 162y - 84y^2 + 204y + 84y - 204 \geq 0$$

$$\Rightarrow -3y^2 + 126y - 123 \geq 0$$

$$\Rightarrow 3y^2 - 126y + 123 \leq 0$$

$$\Rightarrow 3(y^2 - 42y + 41) \leq 0$$

$$\Rightarrow y^2 - 42y + 41 \leq 0 \quad [\because a \cdot b \leq 0, \text{ if } a > 0 \Rightarrow b \leq 0]$$

$$\Rightarrow y^2 - 42y + 41 \leq 0$$

$$\Rightarrow (y-1)(y-41) \leq 0$$

Evaluating all factors to zero, we get

$$y = 1, 41$$

By wave curve method, we have

$$y \in (1, 41)$$

$$\Rightarrow 1 \leq y \leq 41$$

Hence maximum value of y is 41.

Q15 (2) - Range of quadratic expressions

$$\text{Let } y = \frac{(x+m)^2 - 4mn}{2(x-n)}$$

$$\Rightarrow y \cdot 2(x-n) = (x+m)^2 - 4mn$$

$$\Rightarrow 2y(x-n) - (x+m)^2 + 4mn = 0$$

$$\Rightarrow 2yx - 2yn - \{x^2 + m^2 + 2xm\} + 4mn = 0$$

$$\Rightarrow 2yx - 2yn - x^2 - m^2 - 2xm + 4mn = 0$$

$$\Rightarrow -x^2 + 2x(y-m) - 2yn - m^2 + 4mn = 0$$

$$\Rightarrow x^2 - 2x(y-m) + 2yn + m^2 - 4mn = 0$$

which is a quadratic equation in x .

For x to be real, $D \geq 0$

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$$\Rightarrow \{-2(y-m)\}^2 - 4 \times 1 \{2yn + m^2 - 4mn\} \geq 0 \quad [\because D = b^2 - 4ac]$$

$$\Rightarrow 4(y-m)^2 - 4(2yn + m^2 - 4mn) \geq 0$$

$$\Rightarrow 4(y^2 + m^2 - 2my) - 4(2yn + m^2 - 4mn) \geq 0$$

$$\Rightarrow 4\{y^2 + m^2 - 2my - 2yn - m^2 + 4mn\} \geq 0$$

$$\Rightarrow 4\{y^2 - 2y(m+n) + 4mn\} \geq 0$$

$$\Rightarrow y^2 - 2y(m+n) + 4mn \geq 0$$

$$[\because ab \geq 0, \text{ if } a > \Rightarrow b \geq 0]$$

$$\Rightarrow y^2 - 2ym - 2yn + 4mn \geq 0$$

$$\Rightarrow y(y-2m) - 2n(y-2m) \geq 0$$

$$\Rightarrow (y-2m)(y-2n) \geq 0$$

Equating all factors to zero, we get $y = 2m, 2n$

By wavy curve method, we have If $m < n$, then

$y \in (-\infty, 2m] \cup [2n, \infty)$ and if $m > n$, then

$y \in (-\infty, 2n] \cup [2m, \infty)$

Hence, y cannot lie between $2m$ and $2n$.

Q16 (2) - Location of roots

Given equation. $x^2 - 2kx + (k^2 + k - 5) = 0 \dots(i)$

Since, both roots of the Eq. (i) less than 5.

$$(1) D \geq 0$$

$$(2) A \cdot f(5) > 0$$

$$(3) 5 > \frac{-b}{2a}$$

[if both roots of $ax^2 + bx + c = 0$ is less than, k then $D \geq 0, af(k) > 0, k > -b/2a$]

From condition (1), we have $D \geq 0$

$$D \geq 0$$

$$\Rightarrow 163^2 - 4 \times 1 \times (k^2 + k - 5) \geq 0$$

$$\Rightarrow 14^2 - 4(k^2 + k - 5) \geq 0$$

$$\Rightarrow 4[k^2 - k^2 - k + 5] \geq 0$$

$$\Rightarrow -k + 5 \geq 0$$

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$$\Rightarrow k - 5 \leq 0$$

$$\Rightarrow k \leq 5$$

$$\Rightarrow k \in (-\infty, 5] \dots (ii)$$

From condition (2), we have

$$1 \cdot f(5) > 0$$

$$\Rightarrow 25 - 2k \times 5 + (k^2 + k - 5) > 0$$

$$\Rightarrow 25 - 10k + k^2 + k - 5 > 0$$

$$\Rightarrow k^2 - 9k + 20 > 0$$

$$\Rightarrow (k - 5)(k - 4) > 0$$

Equating all factors to zero, we get

$$k = 5, 4$$

By wavy curve method, we have

$$\Rightarrow k \in (-\infty, 4) \cup (5, \infty) \dots (iii)$$

From condition (3), we have

$$5 > -\frac{b}{2a} \Rightarrow 5 > \frac{2k}{2 \times 1}$$

$$5 > k \Rightarrow (-\infty, 5) \dots (iv)$$

Combining Eqs. (ii), (iii) and (iv), we get

$$k \in (-\infty, 4)$$

Q17 (2) - Location of roots

Given equation, $x^2 - 2mx + m^2 - 1 = 0$

Since, both roots of Eq. (i) lie in the interval $(-2, 4)$, then

$$(1) D \geq 0$$

$$(2) A \cdot f(-2) > 0$$

$$(3) A \cdot f(4) > 0$$

$$(4) -2 < \frac{-b}{2a} < 4$$

[$\because$ if both roots of $ax^2 + bx + c = 0$ lies in (k_1, k_2) , then $D \geq 0$, $af(k_1) > 0$, $af(k_2) > 0$ and

$$k_1 < \frac{-b}{2a} < k_2]$$

From condition (1), we have

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$$D \geq 0$$

$$\Rightarrow (-2m)^2 - 4 \times 1 \times (m^2 - 1) \geq 0$$

$$\Rightarrow 4m^2 - 4m^2 + 4 \geq 0$$

$$\Rightarrow 4 \geq 0$$

$$\Rightarrow m \in R \dots (i)$$

From condition (2), we have

$$\Rightarrow 1 \cdot f(-2) > 0$$

$$\Rightarrow (-2)^2 - 2m(-2) + m^2 - 1 > 0$$

$$\Rightarrow 4 + 4m + m^2 - 1 > 0$$

$$\Rightarrow m^2 + 4m + 3 > 0$$

$$\Rightarrow (m + 3)(m + 1) \geq 0$$

Equating all factors to zero, we get

$$m = -1, -3$$

By wavy curve method, we have

$$\Rightarrow m \in (-\infty, -3] \cup [-1, \infty) \dots (ii)$$

$$1 \cdot f(4) > 0$$

$$\Rightarrow 16 - 8m + m^2 - 1 > 0$$

$$\Rightarrow m^2 - 8m + 15 > 0$$

$$\Rightarrow (m - 3)(m - 5) > 0$$

$$\Rightarrow m = 3, 5$$

$$\Rightarrow m \in (-\infty, 3) \cup (5, \infty) \dots (iii)$$

From condition (4) we have

$$-2 < \frac{2m}{2} < 4$$

$$\Rightarrow -2 < m < 4 \dots (iv)$$

Combining in Eas. (i). (ii). (iii) and (iv) we get

$$m \in [-1, 3)$$

Q18 (1) - Location of roots

$$\text{Given equation, } x^2 - 2ax + a^2 + a - 3 = 0 \dots (i)$$

Since, roots are real and less than 3, then

$$(1) D \geq 0$$

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$$(2) A \cdot f(3) > 0$$

$$(3) 3 > \frac{-B}{2A}$$

[$\because$ if roots of $ax^2 + bx + c = 0$ are real and less than k , then $D \geq 0$, $a \cdot f(k) > 0$, $k > -b/2a$]

From condition (1), we have

$$D \geq 0$$

$$\Rightarrow (-2a)^2 - 4(a^2 + a - 3) \geq 0$$

$$\Rightarrow 4a^2 - 4a^2 - 4a + 12 \geq 0$$

$$\Rightarrow -4a + 12 \geq 0$$

$$\Rightarrow 4a - 12 \leq 0$$

$$\Rightarrow 4a \leq 12$$

$$\Rightarrow a \leq 3$$

$$\Rightarrow a \in (-\infty, 3] \dots (ii)$$

From condition (2), we have

$$A \cdot f(3) > 0$$

$$\Rightarrow 1 \cdot f(3) > 0$$

$$\Rightarrow f(3) > 0$$

$$\Rightarrow 9 - 6a + a^2 + a - 3 > 0$$

$$\Rightarrow a^2 - 5a + 6 > 0$$

$$\Rightarrow (a - 3)(a - 2) > 0$$

Equating all factors to zero, we get

$$a = 2, 3$$

$$\Rightarrow a \in (-\infty, 2) \cup (3, \infty) \dots (iii)$$

From condition (3), we have

$$\Rightarrow 3 > \frac{-B}{2A}$$

$$\Rightarrow 3 > \frac{2a}{2}$$

$$\Rightarrow 3 > a$$

$$\Rightarrow a < 3 \dots (iv)$$

$$\Rightarrow a \in (-\infty, 3)$$

Combining Eqs. (ii), (iii) and (iv), we get

$$a \in (-\infty, 2)$$

$$\therefore a < 2$$

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Q19 (1) - Location of roots

We have,

$$x^2 + \left(k + \frac{11}{3}\right)x - (k^2 + k + 1) = 0$$

Product of roots $= -(k^2 + k + 1) < 0$ for all $k \in R$

Hence, one positive and one negative real root, shown as Now,

$$f(0) < 0 \text{ and } f(3) > 0$$

$$\therefore f(3) > 0$$

$$\Rightarrow 9 - \left(k + \frac{11}{3}\right) \cdot 3 - k^2 - k - 1 > 0$$

$$\Rightarrow -k^2 - 4k - 3 > 0$$

$$\Rightarrow k^2 + 4k + 3 < 0$$

$$\therefore k \in (-3, -1)$$

Q20 (4) - Location of roots

Given equation, $5x^2 + (k+1)x + k = 0 \dots(i)$

Since, exactly one root of Eq. (i) lies in $(1, 3)$, then

$$(1) D > 0$$

$$(2) f(1) \cdot f(3) < 0$$

[$\because$ if exactly one root of $ax^2 + bx + c = 0$ lies in (k_1, k_2) , then $D > 0$, $f(k_1) f(k_2) < 0$]

From condition (1), we have

$$D > 0$$

$$\Rightarrow (k+1)^2 - 4 \times 5 \times k > 0$$

$$\Rightarrow k^2 + 1 + 2k - 20k > 0$$

$$\Rightarrow k^2 - 18k + 1 > 0$$

$$\begin{aligned} \Rightarrow k &= \frac{18 \pm \sqrt{(18)^2 - 4}}{2} = \frac{18 \pm \sqrt{324 - 4}}{2} \\ &= \frac{18 \pm \sqrt{320}}{2} = \frac{18 \pm 17.89}{2} \\ &= 17.94, 0.055 \end{aligned}$$

$$\Rightarrow k \in (-\infty, 0.055) \cup (17.94, \infty)$$

From condition (2), we have

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$$\Rightarrow f(1) \cdot f(3) < 0$$

$$\Rightarrow \{(5 + k + 1 + k)(45 + 3k + 3 + k)\} < 0$$

$$\Rightarrow (2k + 6)(4k + 48) < 0$$

Equating all factors to zero, we get

$$k = -3, -12$$

By wavy curve method, we have

$$k \in (-12, -3) \dots (iii)$$

Combining Eqs. (ii) and (iii), we get

$$k \in (-12, -3)$$

Hence, the integral value of k are -4, -5, -6, -7, -8, -9, -10, -11 and total number of integral values of k are 8.